

Who Built the Service Economy?

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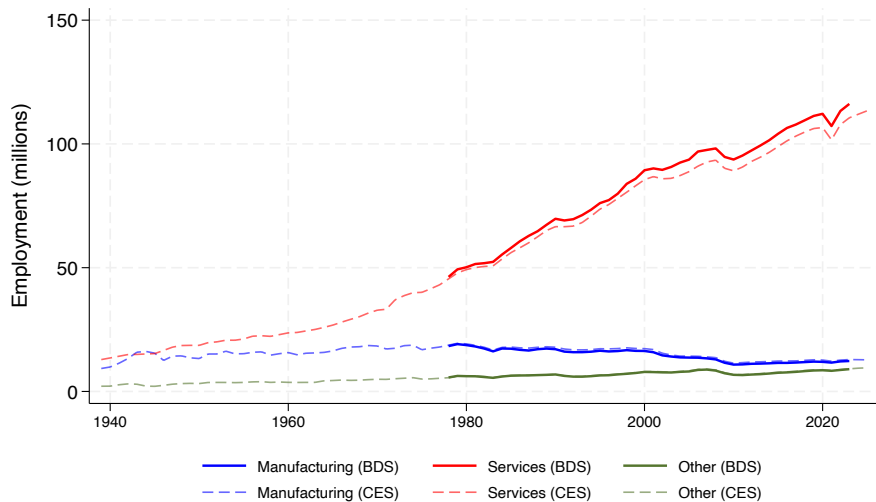
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Sectoral transformation



Our questions

- ▶ How did this large reallocation of labor come about?
- ▶ Was it driven by entry of new service-sector firms?
- ▶ Or by the decline and exit of manufacturing incumbents?
- ▶ Did structural transformation primarily reflect ...
- ▶ ... the rise of small, young service firms?
- ▶ ... or the expansion of large, established incumbents?

We take a **firm-dynamics** perspective: which firms grew, and how?

Data

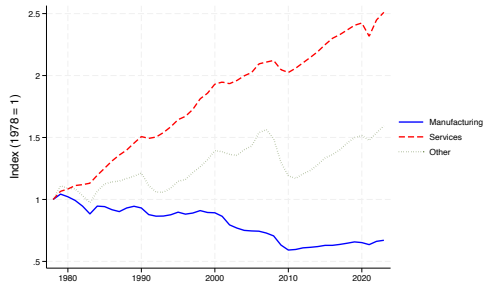
Business Dynamics Statistics (BDS), U.S. Census Bureau, 1978–2023

- built from the LBD: the universe of U.S. establishments with paid employees
- establishment: a single physical location where business is conducted
- firm: all establishments under common ownership
- firm entry: all establishments are new; firm exit: all establishments shut down

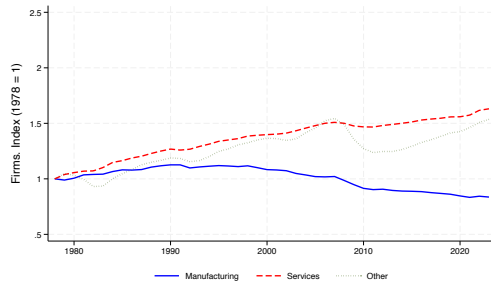
Sectors and firm-size groups

- manufacturing: NAICS 31–33
- services: trade, transportation, information, FIRE, professional, health, . . .
- small firms: 1–19 employees; medium: 20–499; large: 500+

Service employment up 2.5x, firm count only 1.6x

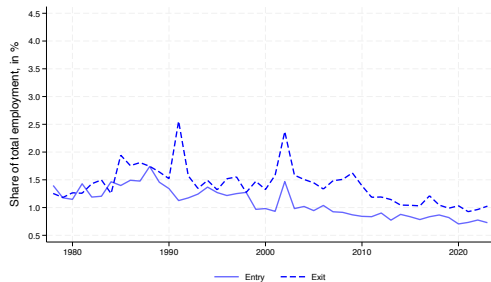


Employment (1978 = 1)

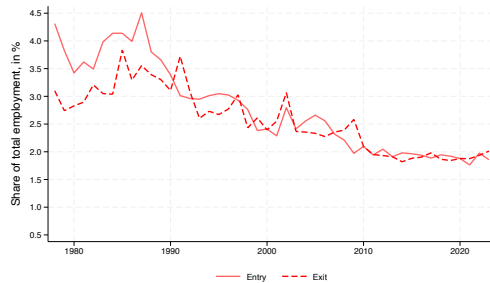


Number of firms (1978 = 1)

Entry and exit play a minor role



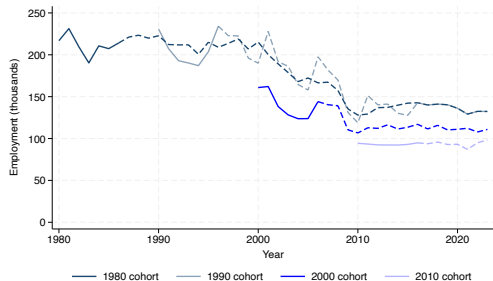
Manufacturing



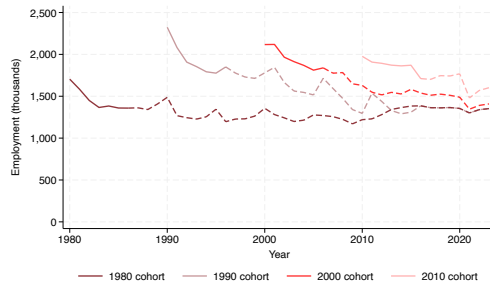
Services

Firm entry and exit rates

Limited role of cohort effects



Manufacturing

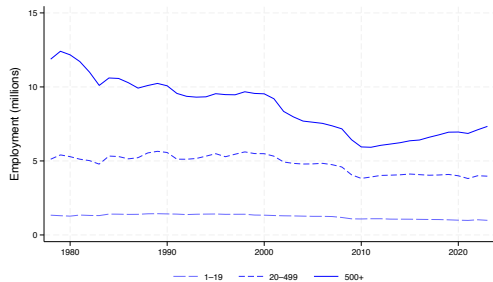


Services

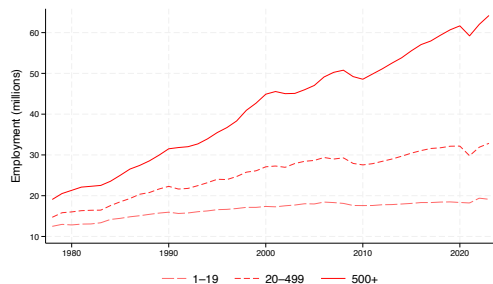
Each line follows all firms born in a given cohort over time.

If entrants' initially small employment contribution grew with age, cohort profiles would be roughly hump-shaped, but we do not see that pattern.

Reallocation is concentrated among large firms

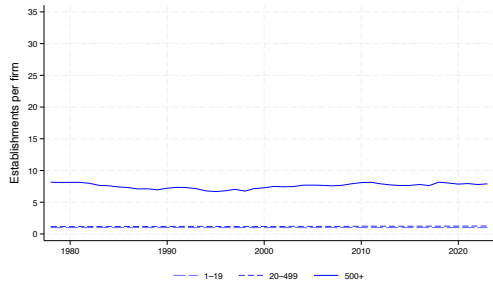


Manufacturing

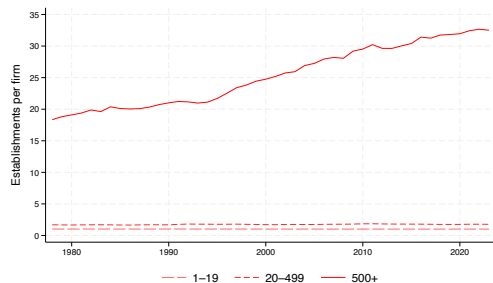


Services

Large service firms grew by adding establishments



Manufacturing



Services

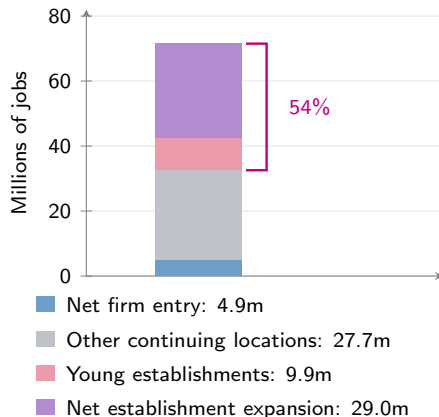
Empirical takeaway: incumbents built the service economy

- ▶ Service employment rose 2.5-fold, number of service firms rose only 1.6-fold.
- ▶ Employment growth was concentrated in large service firms.
- ▶ Large firms expanded their establishment networks, while employment per establishment remained broadly stable.
- ▶ Incumbents account for 93% of service-employment growth.
- ▶ How much of that 93% comes from new establishments?
 - ▶ 41 p.p. arise directly from opening additional establishments
 - ▶ including their measured subsequent growth raises the contribution to over 54 p.p.

New establishments & service job growth

- ▶ Net establishment expansion at incumbent firms added 29.0 million service jobs.
- ▶ Young establishments at mature firms added another measured 9.9 million jobs after opening.
- ▶ This counts net growth at establishments aged 1–5 in firms aged 6+. It does not capture all subsequent growth.

Service employment growth, 1978–2023



But...

- ▶ "...isn't this just health care?" **No.** Service Subsector Levels Service Employment Gains
- ▶ "...did service establishments grow in all service industries?" **Yes.** By Industry
- ▶ "...new establishments, or acquired ones?" **New establishments.** Births
- ▶ "...did establishments get bigger?" **Not in services.** Emp. per Establishment
- ▶ "...what about firm age?" **Similar picture.** Firm Age
- ▶ "...did you formally decompose ΔE ?" **Yes.** Decomposition

Related literature

Firm dynamics

Hopenhayn (1992), Hopenhayn and Rogerson (1993), Klette and Kortum (2004), Luttmer (2007), Bernard et al. (2010), Decker et al. (2014), Hopenhayn et al. (2022), Aghion et al. (2023), Xi (2023), Karahan et al. (2024), Hopenhayn et al. (2025), Peters and Walsh (2026)

- **This paper:** firms can grow through establishments

Structural transformation and multisector growth

Kongsamut et al. (2001), Ngai and Pissarides (2007), Buera and Kaboski (2012), Herrendorf et al. (2014), Herrendorf et al. (2015), Comin et al. (2021)

- **This paper:** adds incumbent establishment decisions to standard ST models

Spatial expansion and multi-establishment firms

Cao et al. (2022), Gumpert et al. (2022), Hsieh and Rossi-Hansberg (2023), Oberfield et al. (2024), Lashkari et al. (2024), Kleinman (2025), Brynjolfsson et al. (2025), Argente et al. (2025)

- **This paper:** quantifies how lower expansion costs reallocate employment across sectors

Model

What we do

We extend Hopenhayn & Rogerson (1993) to two sectors:

- two final-goods sectors $j \in \{M, S\}$, linked by non-homothetic CES demand
- multi-establishment firms with convex **establishment-replication costs**

Two aggregate driving forces (1979 vs. 2023):

- **higher** manufacturing productivity $a_M \rightarrow$ drives sectoral reallocation
- **lower** replication-cost convexity $\eta \rightarrow$ incumbents scale via establishments

What we find:

- two forces explain over 50% of manufacturing employment decline
- standard productivity channel leaves firm policies unchanged
- cheaper replication delivers large-firm, multi-establishment growth
- ... and contributes to ST itself: **1/3** of the model-generated reallocation

Households

Representative household:

- supplies labor $\bar{L}$ inelastically
- income $I_t = W_t \bar{L} + \Pi_t$ (wages plus rebated firm profits)
- maximizes $\mathbb{E}_0 \sum_t \beta^t \ln C_t$ over a CES composite of M & S

Non-homothetic demand:

- a manufacturing subsistence level $\bar{c}_M$ is purchased first
- only the remaining (supernumerary) spending is split across M & S
- as income rises, the manufacturing share of spending falls
- needed so demand shifts toward services, beyond relative prices

Demand equations

Firms and establishments

Each establishment uses a decreasing-returns technology:

$$q = a_j z l^{\alpha_j}, \quad \alpha_j \in (0, 1).$$

A firm runs $m \geq 1$ identical establishments, subject to replication costs:

$$G(m) = c_{mj} \frac{m^{1+\eta_j} - 1}{1 + \eta_j}.$$

Firm flow profit in sector j :

$$\Pi_j(z) = \max_{m \geq 1, l \geq 0} m[p_j a_j z l^{\alpha_j} - Wl] - P \cdot G(m) - P \cdot f_j.$$

Firm dynamics: exit and entry

$$V_j(z) = \Pi_j(z) + \beta(1 - \delta_j) \max\{0, \mathbb{E}[V_j(z') \mid z]\}.$$

- ▶ Conditional on surviving, the firm chooses whether to continue.
- ▶ Baseline: $f_j = 0$, so exit is exogenous; endogenous exit is a robustness exercise.
- ▶ Entrants pay $c_{e,j}$, draw $z \sim \pi_{e,j}$, and enter until:

$$\mathbb{E}_{\pi_{e,j}}[V_j(z)] = Pc_{e,j}.$$

Stationary equilibrium

Extensive margin of firm growth

Establishment-level optimization: decreasing returns ($\alpha_j < 1$)

$$l^*(z) = \left(\frac{p_j \alpha_j a_j z}{W} \right)^{\frac{1}{1-\alpha_j}}, \quad \pi_v(z) \propto z^{\frac{1}{1-\alpha_j}}.$$

Optimal number of establishments per firm:

$$m^*(z) = \max \left\{ 1, \left(\frac{\pi_v(z)}{P_{c_{mj}}} \right)^{1/\eta_j} \right\}.$$

- **Intensive margin**: larger establishments; **extensive margin**: more establishments.
- Only firms with $\pi_v(z) > P_{c_{mj}}$ replicate; convexity η_j sets how fast.

Effective returns to scale

Effective firm-level returns to scale exceed establishment-level returns:

$$L_j(z) = m_j^*(z)l_j^*(z) \propto z^{\frac{1}{1-\alpha_{\text{eff},j}}}, \quad \alpha_{\text{eff},j} = \frac{1 + \eta_j \alpha_j}{1 + \eta_j} \in (\alpha_j, 1).$$

- $\eta_j \rightarrow \infty$: $\alpha_{\text{eff},j} \rightarrow \alpha_j$ (single establishment)
- $\eta_j \rightarrow 0$: $\alpha_{\text{eff},j} \rightarrow 1$ (constant returns)

Analytical results

- ▶ **Proposition 1:** Changes in productivity are offset by price changes, while firm sizes stay fixed.
- ▶ **Proposition 2:** Lower replication-cost convexity η favors establishment growth at more productive firms.

Analytical variant: homothetic demand ($\bar{c}_M = 0$) and labor-denominated costs.

Proposition 1: Invariance to sectoral productivity

For each sector $j \in \{M, S\}$, define $x_j \equiv p_j a_j$. Then:

- (i) ℓ_j^* , m_j^* , profits, V_j , and exit depend on (p_j, a_j) only through x_j
- (ii) free entry uniquely determines x_j : if $a'_j = \lambda a_j$, then $p'_j = p_j / \lambda$
- (iii) the normalized stationary firm-size distribution is invariant to a_j

Intuition:

- higher a_j raises the value of entry, and entry pushes p_j down one for one
- productivity growth moves relative prices and sectoral shares, not firm sizes

Proposition 2: Growth tilts toward productive firms

Conditional on prices (for multi-establishment firms):

$$\log m_j^*(z) = \kappa_j + \frac{1}{\eta_j(1-\alpha_j)} \log z, \quad \frac{\partial \log m_j^*}{\partial \eta_j} < 0, \quad \frac{\partial^2 \log m_j^*}{\partial \eta_j \partial \log z} < 0$$

- lower η_j raises m^* , with a **larger proportional effect** at higher z
- equivalently, lower η_j raises effective returns to scale: $\partial \alpha_{\text{eff},j} / \partial \eta_j < 0$

In general equilibrium:

- lower η_j raises the value of entry, so p_j falls and partly offsets the rise in m^*
- relative prices, demand, and entrant masses adjust: a **quantitative** question

Interpreting a falling η

Changes that made it cheaper to replicate and manage establishments

- Barcodes: larger stores, more productive retail Holmes (2001); Basker (2012).
- Wal-Mart's dense store network saves on distribution costs Holmes (2011).
- Information technology widens managers' span of control Bloom et al. (2014).
- Large business-services firms spend most on IT per worker Eckert et al. (2025).

We capture these changes as a decline in the convexity of replication costs, η .

Quantitative results

Parameterization

Assigned outside the model

- β, ε at standard values; exit rates from BDS, p_S/p_M from BEA

Household

- $\bar{c}_M$ and μ : goods expenditure share and manufacturing employment share

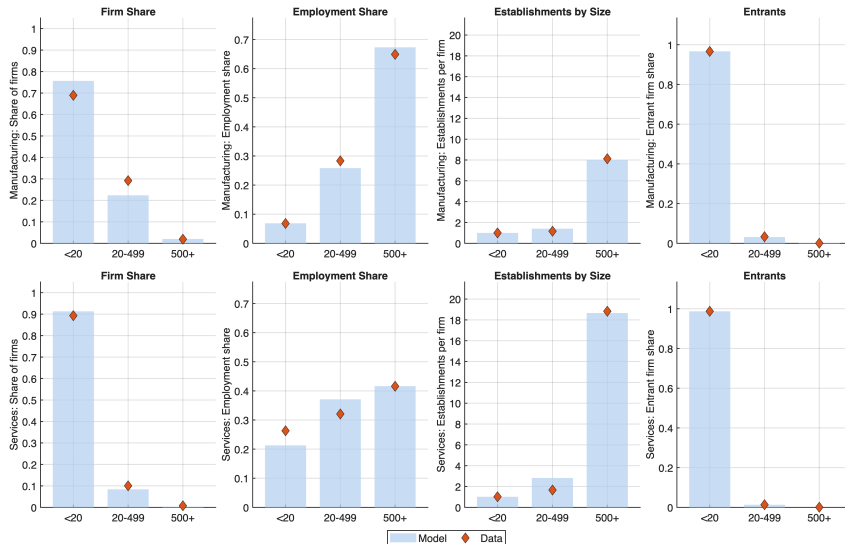
Firms: sector by sector, 1979 BDS moments

- productivity and entrant parameters: firm-size distribution, young-firm growth
- α_j : average establishment size
- $(c_{m,j}, \eta_j)$: establishments per firm, overall and large relative to small

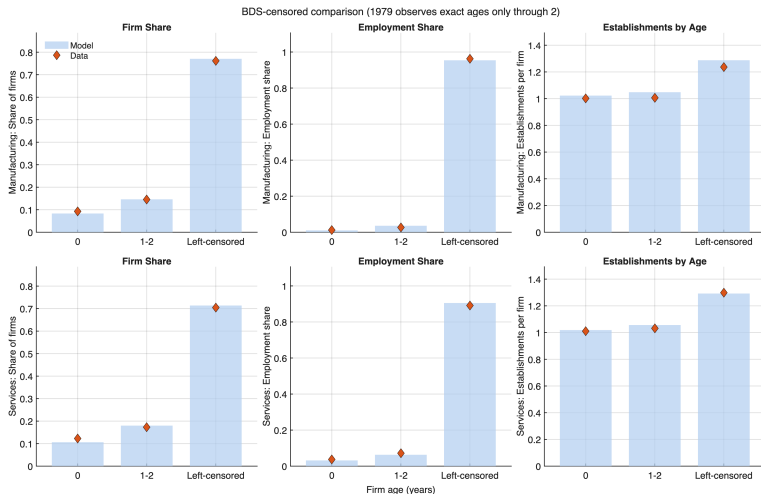
SMM results

All parameters

Initial (1979) steady-state fit



Untargeted fit by firm age



Top: Manufacturing; bottom: Services. In 1979, BDS observes ages 0 and 1–2; ages 3+ are pooled in model and data.

Quantitative experiments

Goal: capture the two forces behind structural transformation

1. manufacturing productivity growth: $a_M \uparrow$
2. organizational change, cheaper replication: $\eta \downarrow$, proportionally in both sectors

From initial 1979 steady state to a new 2023 steady state

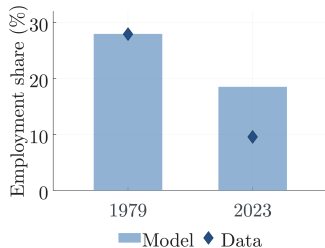
- a_M matches the 1979–2023 change in the relative price of services
- η matches the rise in establishments per firm, large vs. small service firms
- all other primitives fixed; employment share of large service firms untargeted

What we do:

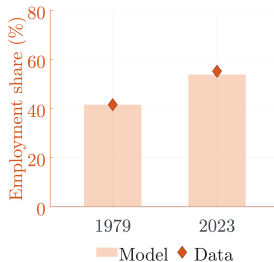
- **Main experiment:** $a_M \uparrow$ and $\eta \downarrow$ together
- **Decomposition:** each force alone, with its corresponding target

Main experiment: $a_M \uparrow$ and $\eta \downarrow$

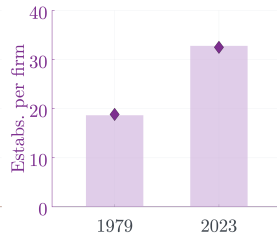
$a_M \times 1.57$, $\eta \times 0.80$ (both sectors): matches the micro facts



Manufacturing employment share

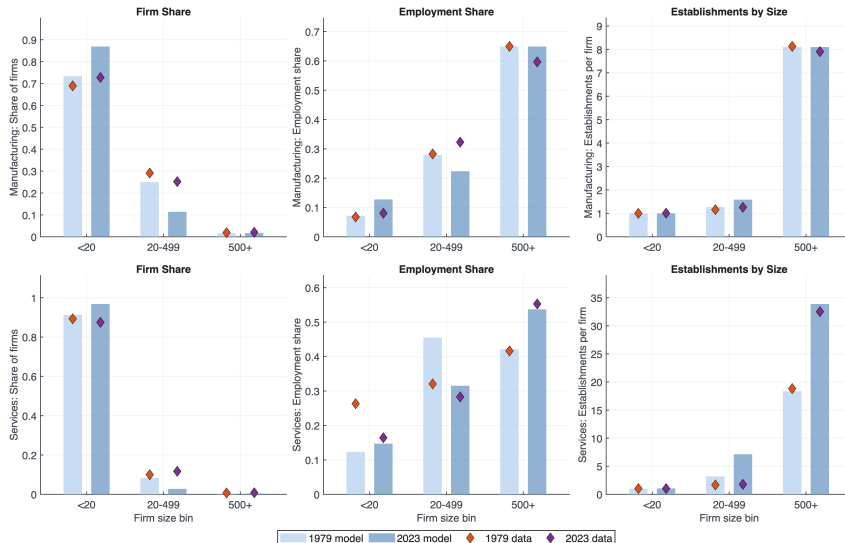


Large (500+) service firms



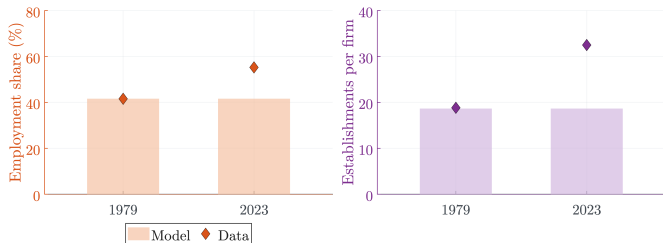
- ✓ generates structural transformation: manufacturing employment share 28.0% → 18.5% (52% of decline)
- ✓ establishments per large service firm 18.7 → 32.8 (targeted)
- ✓ employment share of large service firms 41.6% → 53.9% (untargeted; data 55.3%)

The 2023 firm-size distribution (untargeted)



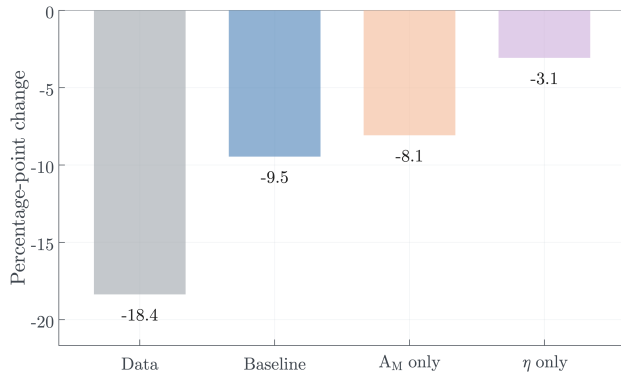
Manufacturing productivity only: $a_M \uparrow$

Raise a_M ($\times 1.55$) to match the manufacturing vs. service relative-price change



- ✓ generates structural transformation: manufacturing employment share 28.0% $\rightarrow$ 19.9% (44% of decline)
- ✗ firm-size distribution does not move: $a_M \uparrow$ is offset by $p_M \downarrow$
- ✗ no shift to large service firms (41.6% vs. 55.3% in data), no establishment growth

Decomposition: Decline in manufacturing employment share



- a_M only: 44% of the decline, but no shift toward large service firms
- η only: 1/3 of joint effect of the baseline model
- the two do not sum to 52%: each triggers its own price adjustment

Model extensions and robustness

How sensitive are the results to the baseline assumptions?

- Growing labor force: higher income shifts demand toward services.
- Endogenous exit: low-productivity firms can choose to shut down.
- Convex entry costs: the supply of entrants is no longer perfectly elastic.

Labor force growth

Endogenous exit

Convex entry costs

Where does a lower η come from? Three recent models

Hsieh & Rossi-Hansberg (2023): units are **markets**

- a firm pays a fixed cost to raise productivity in all of its markets
- firms with more markets gain more, adopt more intensively, and enter still more markets

Brynjolfsson et al. (2025): units are **establishments**

- IT lets a firm run all of its branches the way its best branch runs
- the more branches a firm has, the more it gains, so large chains expand

Argente et al. (2025): units are **products**

- when demand rises, it pays to standardize know-how shared by all products
- productive firms standardize the most, so new products are cheap for them

Isomorphic to a lower η :

- large firms gain more from each change because they have more units
- that is what a lower η does: m^* rises mostly at the top of the distribution

Conclusion

Structural transformation through the lens of firm dynamics

- the reallocation toward services is not driven by entry
- large incumbent service firms expand, mainly along the extensive margin

Implications for modeling

- treating sectors as representative firms misses a key margin of adjustment
- multi-establishment firms: model explains half of the M share decline
- lower replication costs alone deliver 1/3 of the ST the model explains

Who built the service economy?

- existing service firms that expanded establishments across markets

Appendix

References I

- AGHION, P., A. BERGEAUD, T. BOPPART, P. J. KLENOW, AND H. LI (2023): "A Theory of Falling Growth and Rising Rents," *The Review of Economic Studies*, 90, 2675–2702.
- ARGENTE, D., S. MOREIRA, E. OBERFIELD, AND V. VENKATESWARAN (2025): "Scalable Expertise: How Standardization Drives Scale and Scope," NBER Working Paper 34160, National Bureau of Economic Research.
- BASKER, E. (2012): "Raising the Barcode Scanner: Technology and Productivity in the Retail Sector," *American Economic Journal: Applied Economics*, 4, 1–27.
- BERNARD, A. B., S. J. REDDING, AND P. K. SCHOTT (2010): "Multiple-Product Firms and Product Switching," *American Economic Review*, 100, 70–97.
- BLOOM, N., L. GARICANO, R. SADUN, AND J. V. REENEN (2014): "The Distinct Effects of Information Technology and Communication Technology on Firm Organization," *Management Science*, 60, 2859–2885.

References II

- BRYNJOLFSSON, E., W. JIN, G. PETROPOULOS, AND X. WANG (2025): "Information Technology, Firm Size, and Industrial Concentration," NBER Working Paper 31065, National Bureau of Economic Research, first version March 2023.
- BUERA, F. J. AND J. P. KABOSKI (2012): "The Rise of the Service Economy," *American Economic Review*, 102, 2540–69.
- CAO, D., H. R. HYATT, T. MUKOYAMA, AND E. SAGER (2022): "Firm Growth through New Establishments," Tech. rep., Working paper, february 2022 version.
- COMIN, D., D. LASHKARI, AND M. MESTIERI (2021): "Structural Change With Long-Run Income and Price Effects," *Econometrica*, 89, 311–374.
- DECKER, R., J. HALTIWANGER, R. JARMIN, AND J. MIRANDA (2014): "The Role of Entrepreneurship in US Job Creation and Economic Dynamism," *Journal of Economic Perspectives*, 28, 3–24.
- DESMET, K. AND E. ROSSI-HANSBERG (2014): "Spatial Development," *American Economic Review*, 104, 1211–1243.

References III

- DUARTE, M. AND D. RESTUCCIA (2026): “Aggregate Employment and the Rise of Services across Time and Countries,” NBER Working Paper 34804, National Bureau of Economic Research.
- ECKERT, F., S. GANAPATI, AND C. WALSH (2025): “Urban-Biased Growth: A Macroeconomic Analysis,” Working paper, October 2025 revision.
- GUMPERT, A., H. STEIMER, AND M. ANTONI (2022): “Firm Organization with Multiple Establishments,” *The Quarterly Journal of Economics*, 137, 1091–1138.
- HERRENDORF, B., C. HERRINGTON, AND Á. VALENTINYI (2015): “Sectoral Technology and Structural Transformation,” *American Economic Journal: Macroeconomics*, 7, 104–33.
- HERRENDORF, B., R. ROGERSON, AND Á. VALENTINYI (2014): “Growth and Structural Transformation,” in *Handbook of Economic Growth*, ed. by P. Aghion and S. N. Durlauf, Elsevier, vol. 2 of *Handbook of Economic Growth*, 855–941.

References IV

- HOLMES, T. J. (2001): “Bar Codes Lead to Frequent Deliveries and Superstores,” *The RAND Journal of Economics*, 32, 708–725.
- (2011): “The Diffusion of Wal-Mart and Economies of Density,” *Econometrica*, 79, 253–302.
- HOPENHAYN, H., J. NEIRA, AND R. SINGHANIA (2022): “From Population Growth to Firm Demographics: Implications for Concentration, Entrepreneurship and the Labor Share,” *Econometrica*, 90, 1879–1914.
- (2025): “Firm Demographics and Sectoral Reallocation,” Tech. rep., Working paper, december 2025 draft.
- HOPENHAYN, H. AND R. ROGERSON (1993): “Job Turnover and Policy Evaluation: A General Equilibrium Analysis,” *Journal of Political Economy*, 101, 915–938.
- HOPENHAYN, H. A. (1992): “Entry, Exit, and Firm Dynamics in Long Run Equilibrium,” *Econometrica*, 60, 1127–1150.

References V

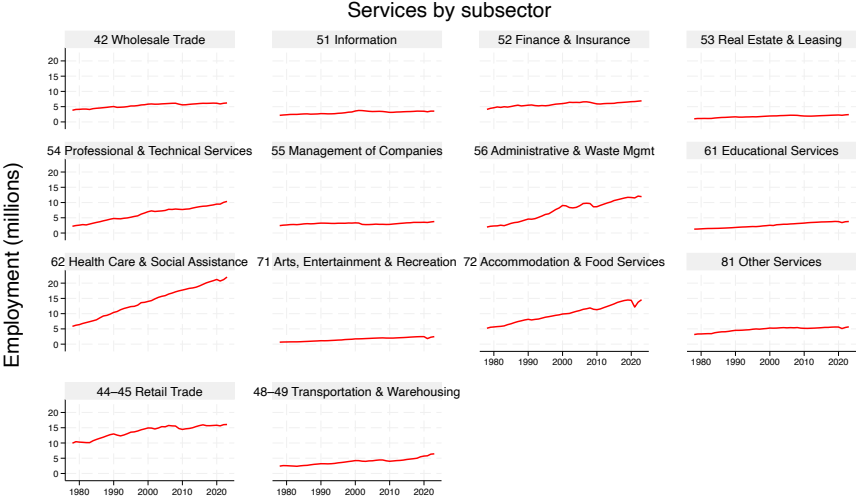
- HSIEH, C.-T. AND E. ROSSI-HANSBERG (2023): “The Industrial Revolution in Services,” *Journal of Political Economy Macroeconomics*, 1, 3–42.
- KARAHAN, F., B. PUGSLEY, AND A. ŞAHİN (2024): “Demographic Origins of the Start-up Deficit,” *American Economic Review*, 114, 1986–2023.
- KLEINMAN, B. (2025): “Wage Inequality and the Spatial Expansion of Firms,” Tech. rep., Stanford University, revise and resubmit, *Quarterly Journal of Economics*.
- KLETTE, T. AND S. KORTUM (2004): “Innovating Firms and Aggregate Innovation,” *Journal of Political Economy*, 112, 986–1018.
- KONGSAMUT, P., S. REBELO, AND D. XIE (2001): “Beyond Balanced Growth,” *The Review of Economic Studies*, 68, 869–882.
- LASHKARI, D., A. BAUER, AND J. BOUSSARD (2024): “Information Technology and Returns to Scale,” *American Economic Review*, 114, 1769–1815.

References VI

- LUTTMER, E. G. J. (2007): "Selection, Growth, and the Size Distribution of Firms," *The Quarterly Journal of Economics*, 122, 1103–1144.
- NGAI, L. R. AND C. A. PISSARIDES (2007): "Structural Change in a Multisector Model of Growth," *American Economic Review*, 97, 429–443.
- OBERFIELD, E., E. ROSSI-HANSBERG, P.-D. SARTE, AND N. TRACHTER (2024): "Plants in Space," *Journal of Political Economy*, 132, 867–909.
- PETERS, M. AND C. WALSH (2026): "Population Growth and Firm Dynamics," *Journal of Political Economy Macroeconomics*, 4, 256–296.
- XI, X. (2023): "Multi-Establishment Firms, Misallocation, and Productivity," *Journal of Economic Dynamics and Control*, 154, 104705.

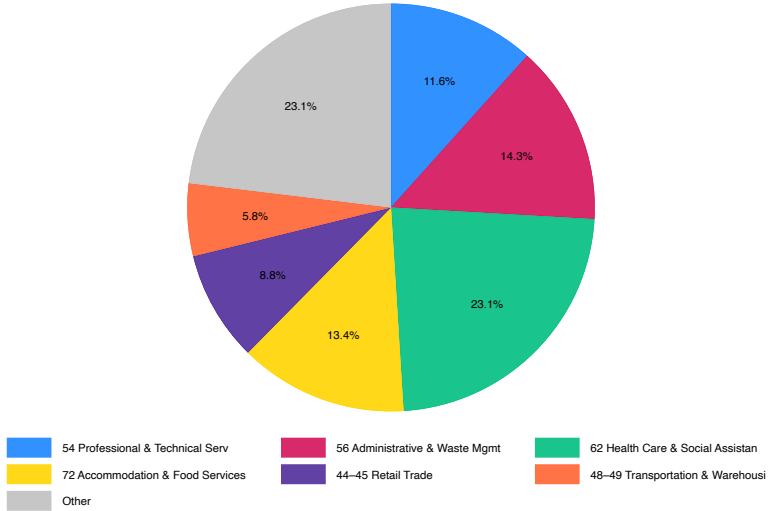
Service Employment by Subsector

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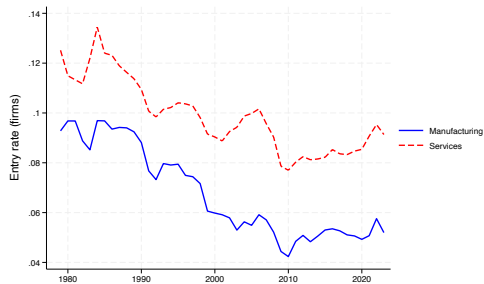


Service Employment Gains by Subsector

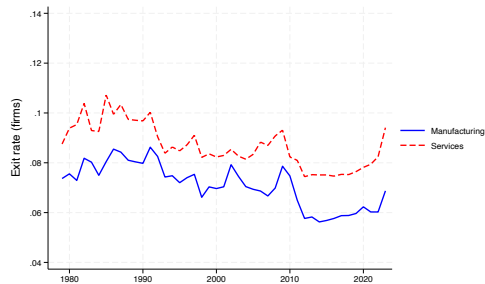
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Entry and Exit Rates

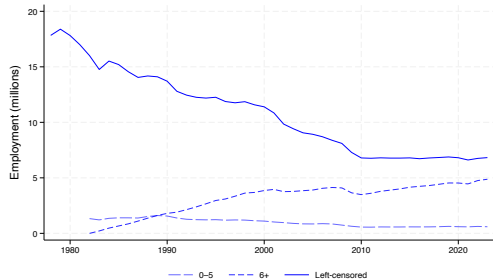
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a) Entry rates

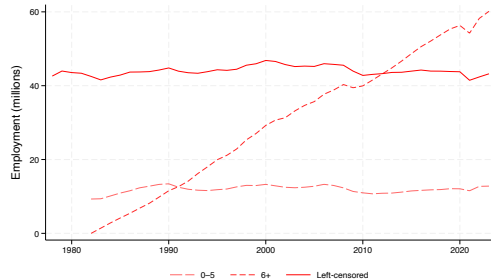


b) Exit rates

Employment by Firm Age

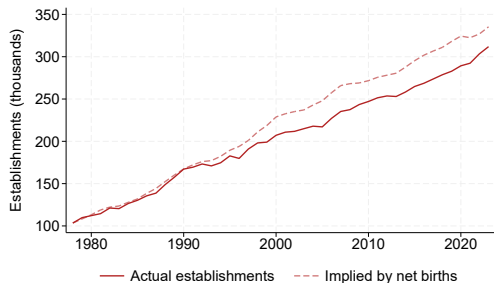
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a) Manufacturing

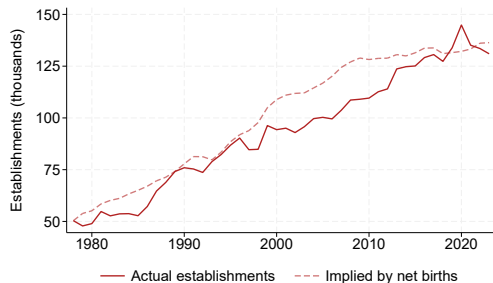


b) Services

New establishments, not acquisitions

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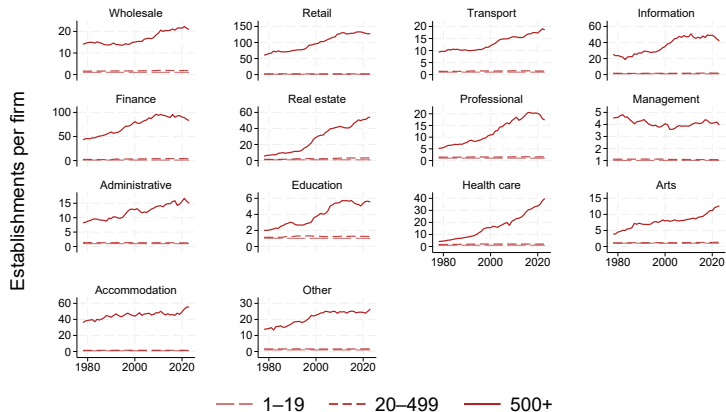
a) Firms with 500–2,499 employees



b) Firms with 2,500–4,999 employees

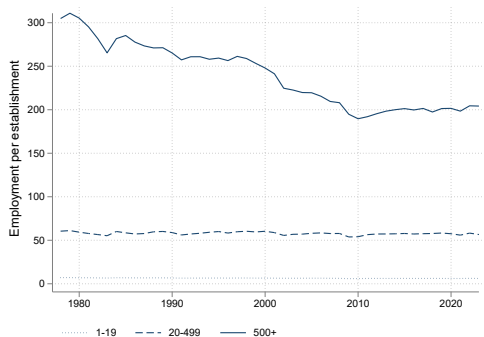
The dashed line adds cumulative net establishment births to the 1978 stock. Its proximity to the observed stock implies that acquisitions play a limited role.

Establishments per firm across service industries

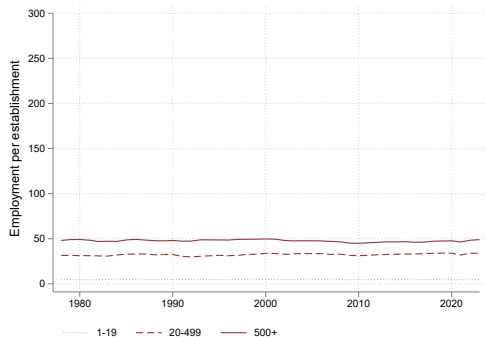
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Firms with 500 or more employees add establishments in nearly every two-digit service industry.

Employment per establishment by firm size

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a) Manufacturing



b) Services

Service establishment size stays stable, so large service firms grow through additional establishments.

Household demand

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Households maximize:

$$U = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \ln(C_t), \quad I_t = W_t \bar{L} + \Pi_t.$$

Manufacturing has a subsistence component:

$$C_{Mt} = \bar{c}_M + \tilde{C}_{Mt}, \quad C_{St} = \tilde{C}_{St}.$$

Supernumerary consumption is aggregated as:

$$C_t = \left[\mu^{\frac{1}{\varepsilon}} \tilde{C}_{Mt}^{\frac{\varepsilon-1}{\varepsilon}} + (1-\mu)^{\frac{1}{\varepsilon}} C_{St}^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}},$$

subject to:

$$P_{Mt} \bar{c}_M + P_t C_t = I_t, \quad P_t = \left[\mu P_{Mt}^{1-\varepsilon} + (1-\mu) P_{St}^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}}.$$

$$\begin{aligned}\Delta E_{s,t} = & \underbrace{\left(JC_{s,t}^{\text{birth}, f=0} - \text{FirmDeathEmp}_{s,t} \right)}_{\text{Net firm entry}} + \underbrace{\left(JC_{s,t}^{\text{cont}} - JD_{s,t}^{\text{cont}} \right)}_{\text{Net within-establishment expansion of incumbents}} \\ & + \underbrace{\left(JC_{s,t}^{\text{birth}, f \geq 1} - \left(JD_{s,t}^{\text{death}} - \text{FirmDeathEmp}_{s,t} \right) \right)}_{\text{Net establishment expansion within incumbent firms}}.\end{aligned}$$

$JC^{\text{birth}, f=0}$: job creation by startups, JC^{cont} , JD^{cont} : job creation/destruction by continuing establishments, $JC^{\text{birth}, f \geq 1}$: new establishments within incumbent firms.

Net entry explains only 7% of service employment growth

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Component	Δ employment (m)	Share (%)
<i>Manufacturing, 1978–2023</i>		
Net firm entry	−2.07	37.7
Net expansion of continuing establishments	−0.60	11.0
Net establishment expansion within incumbents	−2.82	51.3
Total	−5.48	100.0
<i>Services, 1978–2023</i>		
Net firm entry	4.88	6.8
Net expansion of continuing establishments	37.60	52.6
Net establishment expansion within incumbents	29.00	40.6
Total	71.48	100.0

Gross entry and exit flows are large, but mostly offset each other.

Calibration to the 1979 economy

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Parameter	Description	Value	Target	Data	Model
<i>Parameters assigned outside the model</i>					
β	Discount factor	0.96			
ε	Substitution elasticity	0.40	Desmet and Rossi-Hansberg (2014)		
a_M, a_S	Sectoral TFP	1.00	Normalization		
δ_M	Exogenous exit rate	0.08	Average firm entry and exit rate		
δ_S	Exogenous exit rate	0.11	Average firm entry and exit rate		
p_S/p_M	Relative price of services	0.64	BEA value-added price indexes		
<i>Household</i>					
$\bar{c}_M$	Manufacturing subsistence	0.32	Goods-share slope wrt log PCE per capita	-0.23	-0.23
μ	CES preference weight	0.09	Manufacturing employment share	0.28	0.28
<i>Manufacturing</i>					
μ_M^z	Mean log productivity	1.81	Average firm size	73.21	74.16
σ_M	Productivity volatility	0.44	Employment share, firms with 500+	0.65	0.67
$\mu_{e,M}^z$	Mean log entrant productivity	0.27	Average entrant size	9.04	9.03
$\sigma_{e,M}$	Entrant productivity dispersion	0.88	Entrant firm share with 20+	0.03	0.03
ρ_M	Productivity persistence	0.69	Age 1–5 firm share with 20+	0.15	0.15
α_M	Establishment returns to scale	0.53	Average establishment size	61.72	60.21
$c_{m,M}$	Replication-cost level	52.87	Average establishments per firm	1.19	1.23
η_M	Replication-cost convexity	0.87	Estab./firm: 500+ relative to <20	8.10	8.00
$c_{e,M}$	Entry cost	386.56	Free-entry condition	—	—
<i>Services</i>					
μ_S^z	Mean log productivity	0.14	Average firm size	16.43	16.43
σ_S	Productivity volatility	0.18	Employment share, firms with 500+	0.42	0.42
$\mu_{e,S}^z$	Mean log entrant productivity	1.95	Average entrant size	4.90	4.90
$\sigma_{e,S}$	Entrant productivity dispersion	0.24	Entrant firm share with 20+	0.01	0.01
ρ_S	Productivity persistence	0.99	Age 1–5 firm share with 20+	0.06	0.06
α_S	Establishment returns to scale	0.47	Average establishment size	13.46	13.46
$c_{m,S}$	Replication-cost level	14.12	Average establishments per firm	1.22	1.22
η_S	Replication-cost convexity	0.85	Estab./firm: 500+ relative to <20	18.44	18.44
$c_{e,S}$	Entry cost	94.86	Free-entry condition	—	—

Summary of quantitative experiments

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	A_M, η	A_M only	η only	Data
A_M (factor, 1979 = 1)	1.57	1.55	1.00	
Replication-cost scaling	0.80	1.00	0.70	
η_S	0.67	0.85	0.59	
Manufacturing employment share (%)	18.52	19.89	24.90	9.57
Share of decline explained (%)	51.5	44.0	16.7	
Large-firm share of service employment (%)	53.9	41.6	68.9	55.3
Establishments per service firm, 500+	32.8	18.7	32.6	32.5
Average establishments per service firm	1.28	1.22	1.33	1.37

Calibrated firm parameters

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Target		Manufacturing			Services		
		Value	Data	Model	Value	Data	Model
μ_j^z	Average firm size	1.81	73.21	74.16	0.14	16.43	16.43
σ_j	Employment share, firms with 500+	0.44	0.65	0.67	0.18	0.42	0.42
$\mu_{e,j}^z$	Average entrant size	0.27	9.04	9.03	1.95	4.90	4.90
$\sigma_{e,j}$	Entrant firm share with 20+	0.88	0.03	0.03	0.24	0.01	0.01
ρ_j	Age 1–5 firm share with 20+	0.69	0.15	0.15	0.99	0.06	0.06
α_j	Average establishment size	0.53	61.72	60.21	0.47	13.46	13.46
$c_{m,j}$	Average establishments per firm	52.87	1.19	1.23	14.12	1.22	1.22
η_j	Estab./firm: 500+ relative to <20	0.87	8.10	8.00	0.85	18.44	18.44
$c_{e,j}$	Free-entry condition	386.56	—	—	94.86	—	—

Manufacturing: higher α_j and $c_{m,j}$; similar η_j across sectors.

[All parameters](#)

Adding labor-force growth

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Main experiment with labor supply scaled by 1.88 (BDS, 1979–2023)

	Main experiment	+ labor growth	Data
Manufacturing employment share, 2023 (%)	18.5	14.8	9.6
Share of decline explained (%)	51.5	71.8	—
Employment share of large service firms (%)	53.9	53.9	55.3
Establishments per large service firm	32.8	32.8	32.5

Labor-force growth matters for the aggregate, not for the firm-level facts

- higher income shifts spending toward services Duarte & Restuccia (2026)
- the model now accounts for 72% of the manufacturing employment decline
- distributional moments unchanged (free entry)

Endogenous exit

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Recalibrate with both endogenous and exogenous exit

- Exogenous exit: 1.3% in manufacturing, 1.4% in services, from firms with 20+ employees.
- Fixed operating costs match overall entry/exit rates; low-productivity firms choose to exit.

Main experiment, 2023	Baseline	Endogenous exit
Manufacturing employment share (%)	18.5	19.2
Share of observed decline explained (%)	51.5	47.1
Employment share of large service firms (%)	53.9	53.3
Establishments per large service firm	32.8	32.3

The main conclusion survives endogenous selection

Convex entry costs

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Following Karahan et al. (2024), entry costs rise with entrant mass

$$c_{e,j}(M_j) = c_{e,j} \left(\frac{M_j}{M_{j,1979}} \right)^{\psi-1}, \quad \psi \geq 1.$$

- $\psi = 1$: baseline; $\psi > 1$: entrant supply elasticity $1/(\psi - 1)$.
- Re-estimate the shocks using the same targets; initial calibration unchanged.

Manufacturing employment share, 2023 (%)	$\psi = 1$	$\psi = 1.5$	$\psi = 2$
Main experiment	18.5	18.0	18.1
a_M only	19.9	20.2	20.3
η only	24.9	24.8	24.7

The contributions of both channels remain close to baseline

Stationary Equilibrium

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- ▶ Households allocate consumption between manufacturing and service goods.
- ▶ Firm policies solve the establishment, replication, continuation, and exit problems.
- ▶ Free entry holds in each sector:

$$\mathbb{E}_{\pi_{e,j}}[V_j(z)] = P c_{e,j}$$

- ▶ Supernumerary expenditure shares are:

$$\phi_j = \omega_j \left(\frac{p_j}{P} \right)^{1-\varepsilon}, \quad \omega_M = \mu, \quad \omega_S = 1 - \mu$$

- ▶ Goods markets clear, with C_M including $\bar{c}_M$:

$$M_j \int Y_j(z) d\mu_j(z) = C_j + \phi_j B$$

- ▶ Labor market clears:

$$\sum_{j \in \{M, S\}} M_j \int l_j(z) m_j(z) d\mu_j(z) = \bar{L}$$